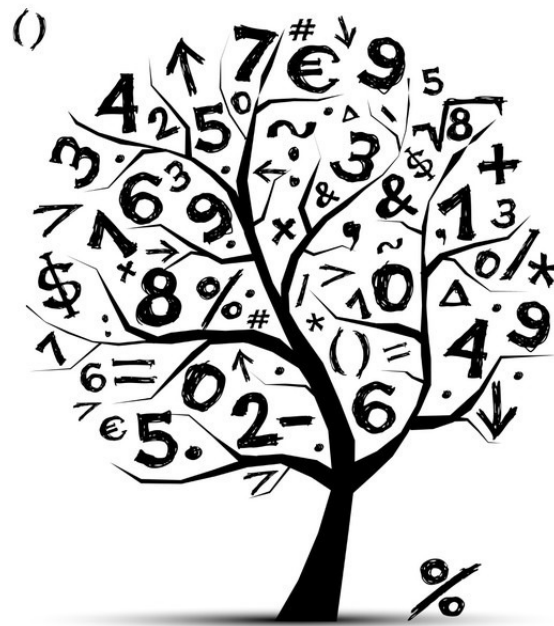


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Exercises: norms



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Exercise 1

Compute the ℓ_1 , ℓ_2 and ℓ_∞ norms of the following column vectors:

$$\mathbf{p} = \begin{pmatrix} 2 \\ -3 \\ 6 \end{pmatrix} \in \mathbb{R}^3, \quad \mathbf{w} = \begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} \in \mathbb{R}^3, \quad \mathbf{u} = \begin{pmatrix} 1 \\ -1 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^4$$

For each vector, check that $\|\cdot\|_\infty \leq \|\cdot\|_2 \leq \|\cdot\|_1$.

Using the definitions, we have:

$$\|\mathbf{p}\|_1 = |2| + |-3| + |6| = 11, \quad \|\mathbf{p}\|_2 = \sqrt{4 + 9 + 36} = 7, \quad \|\mathbf{p}\|_\infty = \max\{2, 3, 6\} = 6$$

$$\|\mathbf{w}\|_1 = |4| + |0| + |-3| = 7, \quad \|\mathbf{w}\|_2 = \sqrt{16 + 0 + 9} = 5, \quad \|\mathbf{w}\|_\infty = \max\{4, 0, 3\} = 4$$

$$\|\mathbf{u}\|_1 = 1 + 1 + 1 + 1 = 4, \quad \|\mathbf{u}\|_2 = \sqrt{1 + 1 + 1 + 1} = 2, \quad \|\mathbf{u}\|_\infty = \max\{1, 1, 1, 1\} = 1$$

Accordingly:

$$6 \leq 7 \leq 11, \quad 4 \leq 5 \leq 7, \quad 1 \leq 2 \leq 4$$

Exercise 2

Consider the column vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \in \mathbb{R}^3 \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix} \in \mathbb{R}^3$$

Compute the distance between $\mathbf{p}$ and $\mathbf{w}$ measured with the ℓ_1 , ℓ_2 and ℓ_∞ norms, i.e., $\|\mathbf{p} - \mathbf{w}\|_1$, $\|\mathbf{p} - \mathbf{w}\|_2$ and $\|\mathbf{p} - \mathbf{w}\|_\infty$. Then explain why $\|\mathbf{w} - \mathbf{p}\| = \|\mathbf{p} - \mathbf{w}\|$ for every norm.

The difference of the two vectors is

$$\mathbf{p} - \mathbf{w} = \begin{pmatrix} 1 - 3 \\ 2 - (-1) \\ -1 - 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \\ -6 \end{pmatrix}$$

hence:

$$\|\mathbf{p} - \mathbf{w}\|_1 = 2 + 3 + 6 = 11, \quad \|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{4 + 9 + 36} = 7, \quad \|\mathbf{p} - \mathbf{w}\|_\infty = \max\{2, 3, 6\} = 6$$

Since $\mathbf{w} - \mathbf{p} = (-1)(\mathbf{p} - \mathbf{w})$, by absolute homogeneity (with $\lambda = -1$) we get, for every norm,

$$\|\mathbf{w} - \mathbf{p}\| = |-1| \|\mathbf{p} - \mathbf{w}\| = \|\mathbf{p} - \mathbf{w}\|$$

Exercise 3

Consider the column vector

$$\mathbf{p} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} \in \mathbb{R}^3$$

1. Find the vector $\mathbf{u}$ with the same direction and orientation as $\mathbf{p}$ such that $\|\mathbf{u}\|_2 = 1$ (i.e., normalize $\mathbf{p}$ with respect to the ℓ_2 norm), and check the result.
2. Normalize $\mathbf{p}$ with respect to the ℓ_1 norm and with respect to the ℓ_∞ norm.
3. Explain why the zero vector cannot be normalized.

1. We have $\|\mathbf{p}\|_2 = \sqrt{4+1+4} = 3$. For any norm, the vector $\mathbf{u} = \frac{1}{\|\mathbf{p}\|} \mathbf{p}$ has, by absolute homogeneity, $\|\mathbf{u}\| = \frac{1}{\|\mathbf{p}\|} \|\mathbf{p}\| = 1$. Hence

$$\mathbf{u} = \frac{1}{3} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{2}{3} \\ -\frac{1}{3} \\ \frac{2}{3} \end{pmatrix}, \quad \|\mathbf{u}\|_2 = \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{4}{9}} = \sqrt{1} = 1$$

2. We have $\|\mathbf{p}\|_1 = 2 + 1 + 2 = 5$ and $\|\mathbf{p}\|_\infty = 2$, hence

$$\frac{1}{5} \mathbf{p} = \begin{pmatrix} \frac{2}{5} \\ -\frac{1}{5} \\ \frac{2}{5} \end{pmatrix}, \quad \left\| \frac{1}{5} \mathbf{p} \right\|_1 = \frac{2}{5} + \frac{1}{5} + \frac{2}{5} = 1, \quad \frac{1}{2} \mathbf{p} = \begin{pmatrix} 1 \\ -\frac{1}{2} \\ 1 \end{pmatrix}, \quad \left\| \frac{1}{2} \mathbf{p} \right\|_\infty = 1$$

3. By definiteness, $\|\mathbf{0}\| = 0$, so we cannot divide by it. Moreover, $\|\lambda \mathbf{0}\| = \|\mathbf{0}\| = 0 \neq 1$ for every $\lambda \in \mathbb{R}$.

Exercise 4

Consider the column vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \in \mathbb{R}^3 \quad \text{and} \quad \mathbf{w} = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix} \in \mathbb{R}^3$$

1. Verify the triangle inequality $\|\mathbf{p} + \mathbf{w}\| \leq \|\mathbf{p}\| + \|\mathbf{w}\|$ for the ℓ_1 , ℓ_2 and ℓ_∞ norms.
2. Verify the reverse triangle inequality $|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| \leq \|\mathbf{p} - \mathbf{w}\|_2$.

We have

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{p} - \mathbf{w} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$$

1.
 - ℓ_1 : $\|\mathbf{p}\|_1 = 5$, $\|\mathbf{w}\|_1 = 5$, $\|\mathbf{p} + \mathbf{w}\|_1 = 6$, and $6 \leq 5 + 5 = 10$.
 - ℓ_2 : $\|\mathbf{p}\|_2 = \sqrt{1+4+4} = 3$, $\|\mathbf{w}\|_2 = \sqrt{4+4+1} = 3$, $\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{9+0+9} = 3\sqrt{2} \approx 4.243$, and $3\sqrt{2} \leq 3 + 3 = 6$.
 - ℓ_∞ : $\|\mathbf{p}\|_\infty = 2$, $\|\mathbf{w}\|_\infty = 2$, $\|\mathbf{p} + \mathbf{w}\|_\infty = 3$, and $3 \leq 2 + 2 = 4$.
2. $\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{1+16+1} = \sqrt{18} = 3\sqrt{2}$, and

$$|\|\mathbf{p}\|_2 - \|\mathbf{w}\|_2| = |3 - 3| = 0 \leq 3\sqrt{2}$$

Exercise 5

Consider the column vectors

$$\mathbf{p} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w} = \begin{pmatrix} 4 \\ -5 \\ 6 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -2 \\ -4 \\ -6 \end{pmatrix}$$

Verify the Cauchy–Schwarz inequality $|\mathbf{p}'\mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$ for the pair $\mathbf{p}, \mathbf{w}$ and for the pair $\mathbf{p}, \mathbf{u}$. In which case does equality hold?

We have $\|\mathbf{p}\|_2 = \sqrt{1+4+9} = \sqrt{14}$.

- Pair $\mathbf{p}, \mathbf{w}$: $\mathbf{p}'\mathbf{w} = 4 - 10 + 18 = 12$ and $\|\mathbf{w}\|_2 = \sqrt{16+25+36} = \sqrt{77}$, hence

$$|\mathbf{p}'\mathbf{w}| = 12 \leq \sqrt{14} \sqrt{77} = \sqrt{1078} \approx 32.83$$

The inequality is strict.

- Pair $\mathbf{p}, \mathbf{u}$: $\mathbf{p}'\mathbf{u} = -2 - 8 - 18 = -28$ and $\|\mathbf{u}\|_2 = \sqrt{4+16+36} = \sqrt{56} = 2\sqrt{14}$, hence

$$|\mathbf{p}'\mathbf{u}| = 28 = \sqrt{14} \cdot 2\sqrt{14} = \|\mathbf{p}\|_2 \|\mathbf{u}\|_2$$

Equality holds: indeed $\mathbf{u} = -2\mathbf{p}$, i.e., $\mathbf{u}$ is a scalar multiple of $\mathbf{p}$.

Exercise 6

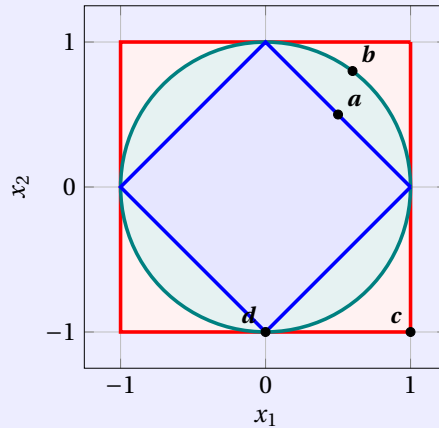
Consider the points of $\mathbb{R}^2$

$$\mathbf{a} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$$



1. Describe and sketch the unit balls $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\| \leq 1\}$ of the ℓ_1 , ℓ_2 and ℓ_∞ norms.
2. For each point, establish whether it lies inside, on the boundary, or outside each of the three unit balls.

1. The ℓ_1 unit ball $|x_1| + |x_2| \leq 1$ is a diamond (a square rotated by 45°) with vertices $(\pm 1, 0)$ and $(0, \pm 1)$; the ℓ_2 unit ball $x_1^2 + x_2^2 \leq 1$ is the disk bounded by the unit circle; the ℓ_∞ unit ball $\max\{|x_1|, |x_2|\} \leq 1$ is the axis-aligned square with vertices $(\pm 1, \pm 1)$.



2. We compute the three norms of each point (value < 1 : inside; $= 1$: on the boundary; > 1 : outside):

	ℓ_1	ℓ_2	ℓ_∞
a	1 (boundary)	$\frac{\sqrt{2}}{2} \approx 0.707$ (inside)	$\frac{1}{2}$ (inside)
b	$\frac{7}{5}$ (outside)	$\sqrt{\frac{9}{25} + \frac{16}{25}} = 1$ (boundary)	$\frac{4}{5}$ (inside)
c	2 (outside)	$\sqrt{2}$ (outside)	1 (boundary)
d	1 (boundary)	1 (boundary)	1 (boundary)

Exercise 7

Consider the matrix

$$\mathbf{Q} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix} \in \mathbb{R}^{2 \times 2}$$

1. Using the definition, prove that $\mathbf{Q}$ is positive definite, so that $\|\mathbf{x}\|_{\mathbf{Q}} = \sqrt{\mathbf{x}' \mathbf{Q} \mathbf{x}}$ is a norm on $\mathbb{R}^2$.

2. Compute $\|\mathbf{p}\|_Q$, $\|\mathbf{w}\|_Q$ and $\|\mathbf{p} + \mathbf{w}\|_Q$ for $\mathbf{p} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.
3. Verify the triangle inequality and the generalized Cauchy–Schwarz inequality for $\mathbf{p}$ and $\mathbf{w}$.

1. $\mathbf{Q}$ is symmetric and, for every $\mathbf{x} \in \mathbb{R}^2$,

$$\mathbf{x}' \mathbf{Q} \mathbf{x} = 5x_1^2 + 4x_1x_2 + 2x_2^2 = x_1^2 + (2x_1 + x_2)^2 + x_2^2 \geq 0$$

The sum of squares is zero only if $x_1 = 0$ and $x_2 = 0$; hence $\mathbf{x}' \mathbf{Q} \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$, i.e., $\mathbf{Q}$ is positive definite, and $\|\cdot\|_Q$ is a generalized ℓ_2 norm.

2. Using $\|\mathbf{x}\|_Q^2 = 5x_1^2 + 4x_1x_2 + 2x_2^2$:

$$\|\mathbf{p}\|_Q = \sqrt{5}, \quad \|\mathbf{w}\|_Q = \sqrt{2}, \quad \|\mathbf{p} + \mathbf{w}\|_Q = \left\| \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\|_Q = \sqrt{5 + 4 + 2} = \sqrt{11}$$

3. Triangle inequality: $\sqrt{11} \approx 3.317 \leq \sqrt{5} + \sqrt{2} \approx 2.236 + 1.414 = 3.650$.

Generalized Cauchy–Schwarz inequality: $\mathbf{p}' \mathbf{Q} \mathbf{w} = q_{12} = 2$, and

$$|\mathbf{p}' \mathbf{Q} \mathbf{w}| = 2 \leq \sqrt{5} \sqrt{2} = \sqrt{10} \approx 3.162$$

Note that in the standard ℓ_2 norm the same vectors are orthogonal ($\mathbf{p}' \mathbf{w} = 0$), while $\mathbf{p}' \mathbf{Q} \mathbf{w} \neq 0$.

Exercise 8

Consider again the symmetric positive definite matrix $\mathbf{Q} = \begin{pmatrix} 5 & 2 \\ 2 & 2 \end{pmatrix}$.

1. Compute the eigenvalues and the eigenvectors of $\mathbf{Q}$.
2. Describe the ellipse $\{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x}' \mathbf{Q} \mathbf{x} = 1\}$: its semi-axes (lengths and directions), the endpoints of the semi-axes, and its intersections with the coordinate axes. Sketch the set $\{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_Q \leq 1\}$.

1. The characteristic equation is

$$\det(\mathbf{Q} - \lambda \mathbf{I}) = (5 - \lambda)(2 - \lambda) - 4 = \lambda^2 - 7\lambda + 6 = (\lambda - 1)(\lambda - 6) = 0$$

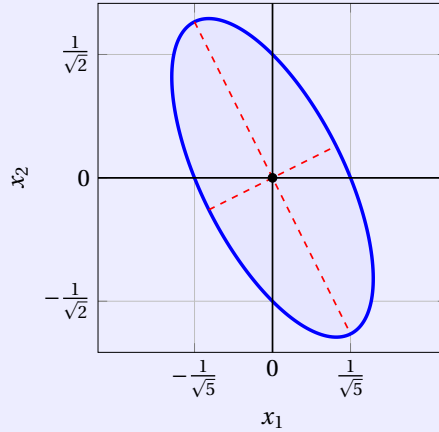
hence $\lambda_1 = 1$ and $\lambda_2 = 6$ (both positive, as expected). The corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \quad \left(\mathbf{Q} \mathbf{v}_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix} = 1 \cdot \mathbf{v}_1 \right), \quad \mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad \left(\mathbf{Q} \mathbf{v}_2 = \begin{pmatrix} 12 \\ 6 \end{pmatrix} = 6 \mathbf{v}_2 \right)$$

2. The ellipse $5x_1^2 + 4x_1x_2 + 2x_2^2 = 1$ is centered at the origin and rotated (its principal axes are the directions of $\mathbf{v}_1$ and $\mathbf{v}_2$, not the coordinate axes).

- Semi-axis along $\mathbf{v}_1$: length $\frac{1}{\sqrt{\lambda_1}} = 1$, endpoints $\pm \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|_2} = \pm \left(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right)$.
- Semi-axis along $\mathbf{v}_2$: length $\frac{1}{\sqrt{\lambda_2}} = \frac{1}{\sqrt{6}}$, endpoints $\pm \frac{1}{\sqrt{6}} \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|_2} = \pm \left(\frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}} \right)$.
- Intersections with the axes: for $x_2 = 0$, $5x_1^2 = 1$, i.e., $\left(\pm \frac{1}{\sqrt{5}}, 0 \right)$; for $x_1 = 0$, $2x_2^2 = 1$, i.e., $\left(0, \pm \frac{1}{\sqrt{2}} \right)$.

For instance, $\left(\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}} \right)$ lies on the ellipse: $5 \cdot \frac{1}{5} + 4 \cdot \left(-\frac{2}{5} \right) + 2 \cdot \frac{4}{5} = 1$.



Exercise 9

Prove that, for every column vector $\mathbf{x} \in \mathbb{R}^n$,

$$\|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty \quad \text{and} \quad \|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_1$$

For which vectors does equality hold in each inequality?

- Since $x_j^2 = |x_j|^2 \leq \|\mathbf{x}\|_\infty^2$ for all $j \in \{1, 2, \dots, n\}$, we have

$$\|\mathbf{x}\|_2^2 = \sum_{j=1}^n x_j^2 \leq \sum_{j=1}^n \|\mathbf{x}\|_\infty^2 = n \|\mathbf{x}\|_\infty^2$$

and, taking the square root of both (non-negative) sides, $\|\mathbf{x}\|_2 \leq \sqrt{n} \|\mathbf{x}\|_\infty$. Equality holds if and only if $x_j^2 = \|\mathbf{x}\|_\infty^2$ for all j , i.e., all components have the same absolute value (e.g., $\mathbf{x} = (1, -1, 1, -1)'$ of Exercise 1: $2 = \sqrt{4} \cdot 1$).

- Let k be an index such that $|x_k| = \|\mathbf{x}\|_\infty$. Then

$$\|\mathbf{x}\|_\infty = |x_k| \leq |x_k| + \sum_{j \neq k} |x_j| = \|\mathbf{x}\|_1$$

Equality holds if and only if $\sum_{j \neq k} |x_j| = 0$, i.e., $\mathbf{x}$ has at most one non-zero component.

Exercise 10

Let $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ with $\mathbf{w} \neq \mathbf{0}$. Prove that equality holds in the Cauchy–Schwarz inequality, i.e.,

$$|\mathbf{p}' \mathbf{w}| = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$$

if and only if $\mathbf{p} = \lambda \mathbf{w}$ for some scalar $\lambda \in \mathbb{R}$.

Hint: use the vector $\mathbf{u} = \alpha \mathbf{p} + \beta \mathbf{w}$, with $\alpha = \mathbf{w}' \mathbf{w}$ and $\beta = -\mathbf{p}' \mathbf{w}$, introduced in the proof of the Cauchy–Schwarz inequality.

($\Leftarrow$) If $\mathbf{p} = \lambda \mathbf{w}$, then

$$|\mathbf{p}' \mathbf{w}| = |\lambda \mathbf{w}' \mathbf{w}| = |\lambda| \|\mathbf{w}\|_2^2 \quad \text{and} \quad \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 = |\lambda| \|\mathbf{w}\|_2 \|\mathbf{w}\|_2 = |\lambda| \|\mathbf{w}\|_2^2$$

($\Rightarrow$) As shown in the proof of the Cauchy–Schwarz inequality, with $\alpha = \mathbf{w}' \mathbf{w}$ and $\beta = -\mathbf{p}' \mathbf{w}$ we have

$$\|\mathbf{u}\|_2^2 = \mathbf{u}' \mathbf{u} = \mathbf{w}' \mathbf{w} ((\mathbf{w}' \mathbf{w}) (\mathbf{p}' \mathbf{p}) - (\mathbf{p}' \mathbf{w})^2)$$

If $|\mathbf{p}' \mathbf{w}| = \|\mathbf{p}\|_2 \|\mathbf{w}\|_2$, then $(\mathbf{p}' \mathbf{w})^2 = (\mathbf{p}' \mathbf{p}) (\mathbf{w}' \mathbf{w})$, so the term in parentheses is zero and $\|\mathbf{u}\|_2 = 0$. By definiteness, $\mathbf{u} = \mathbf{0}$, i.e.,

$$(\mathbf{w}' \mathbf{w}) \mathbf{p} - (\mathbf{p}' \mathbf{w}) \mathbf{w} = \mathbf{0} \quad \Rightarrow \quad \mathbf{p} = \lambda \mathbf{w} \quad \text{with} \quad \lambda = \frac{\mathbf{p}' \mathbf{w}}{\mathbf{w}' \mathbf{w}}$$

where we divided by $\mathbf{w}' \mathbf{w} = \|\mathbf{w}\|_2^2 > 0$.

For instance, in Exercise 5, $\mathbf{u} = -2\mathbf{p}$ and equality holds, while $\mathbf{w}$ is not a multiple of $\mathbf{p}$ and the inequality is strict.

Exercise 11

For $\mathbf{x} \in \mathbb{R}^2$, consider the functions

$$f(\mathbf{x}) = |x_1| + 2|x_2|, \quad g(\mathbf{x}) = |x_1|, \quad h(\mathbf{x}) = x_1^2 + x_2^2$$

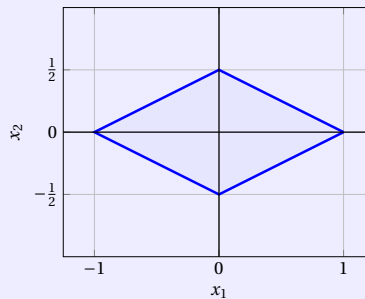
1. Prove that f is a norm and sketch the set $\{\mathbf{x} \in \mathbb{R}^2 : f(\mathbf{x}) \leq 1\}$.
2. Show, with a counterexample, that g and h are not norms.

1. We check the three properties.

- Non-negativity and definiteness: $f(\mathbf{x}) \geq 0$ as a sum of non-negative terms; $f(\mathbf{x}) = 0$ if and only if $|x_1| = 0$ and $|x_2| = 0$, i.e., $\mathbf{x} = \mathbf{0}$.
- Absolute homogeneity: $f(\lambda \mathbf{x}) = |\lambda x_1| + 2|\lambda x_2| = |\lambda|(|x_1| + 2|x_2|) = |\lambda|f(\mathbf{x})$.
- Triangle inequality: by the triangle inequality for the absolute value,

$$f(\mathbf{x} + \mathbf{y}) = |x_1 + y_1| + 2|x_2 + y_2| \leq |x_1| + |y_1| + 2|x_2| + 2|y_2| = f(\mathbf{x}) + f(\mathbf{y})$$

The set $|x_1| + 2|x_2| \leq 1$ is a diamond with vertices $(\pm 1, 0)$ and $(0, \pm \frac{1}{2})$:



2. g violates definiteness: $g((0, 1)') = 0$ but $(0, 1)' \neq \mathbf{0}$.

h violates absolute homogeneity: for $\mathbf{x} = (1, 0)'$ and $\lambda = 2$, $h(2\mathbf{x}) = 4 \neq 2 = |\lambda| h(\mathbf{x})$. It also violates the triangle inequality: $h((1, 0)' + (1, 0)') = 4 > 1 + 1$. (Note that $h(\mathbf{x}) = \|\mathbf{x}\|_2^2$: the square of a norm is not a norm.)

Exercise 12

Let $\mathbf{p}, \mathbf{w} \in \mathbb{R}^n$ be such that $\|\mathbf{p}\|_2 = 3$ and $\|\mathbf{w}\|_2 = 4$.

1. Find the smallest and the largest possible values of $\|\mathbf{p} + \mathbf{w}\|_2$, and give vectors attaining them.
2. Compute $\|\mathbf{p} + \mathbf{w}\|_2$ if $\mathbf{p}'\mathbf{w} = 0$.
3. Compute $\|\mathbf{p} + \mathbf{w}\|_2$ and $\|\mathbf{p} - \mathbf{w}\|_2$ if $\mathbf{p}'\mathbf{w} = 6$.

4. Is it possible that $\mathbf{p}'\mathbf{w} = 13$?

1. By the triangle inequality, $\|\mathbf{p} + \mathbf{w}\|_2 \leq 3 + 4 = 7$. By the reverse triangle inequality applied to $\mathbf{p}$ and $-\mathbf{w}$ (note that $\|-\mathbf{w}\|_2 = \|\mathbf{w}\|_2$),

$$\|\mathbf{p} + \mathbf{w}\|_2 = \|\mathbf{p} - (-\mathbf{w})\|_2 \geq \left| \|\mathbf{p}\|_2 - \|\mathbf{w}\|_2 \right| = |3 - 4| = 1$$

Both bounds are attained: if $\mathbf{w} = \frac{4}{3}\mathbf{p}$, then $\|\mathbf{p} + \mathbf{w}\|_2 = \frac{7}{3} \cdot 3 = 7$; if $\mathbf{w} = -\frac{4}{3}\mathbf{p}$, then $\|\mathbf{p} + \mathbf{w}\|_2 = \frac{1}{3} \cdot 3 = 1$. For instance, $\mathbf{p} = (3, 0)'$ and $\mathbf{w} = (\pm 4, 0)'$.

2. Using $\|\mathbf{p} + \mathbf{w}\|_2^2 = \|\mathbf{p}\|_2^2 + \|\mathbf{w}\|_2^2 + 2\mathbf{p}'\mathbf{w}$, we get $\|\mathbf{p} + \mathbf{w}\|_2^2 = 9 + 16 + 0 = 25$, hence $\|\mathbf{p} + \mathbf{w}\|_2 = 5$ (Pythagorean theorem).

3. $\|\mathbf{p} + \mathbf{w}\|_2^2 = 9 + 16 + 12 = 37$ and $\|\mathbf{p} - \mathbf{w}\|_2^2 = 9 + 16 - 12 = 13$, hence $\|\mathbf{p} + \mathbf{w}\|_2 = \sqrt{37}$ and $\|\mathbf{p} - \mathbf{w}\|_2 = \sqrt{13}$.

4. No: by the Cauchy–Schwarz inequality, $|\mathbf{p}'\mathbf{w}| \leq \|\mathbf{p}\|_2 \|\mathbf{w}\|_2 = 12 < 13$.